Week 15 — Extended Reading: DT, Trajectory Transformer, Diffusion Policies

STUDY-RL

May 21, 2026

1 Chen et al. (2021) — Decision Transformer

L. Chen, K. Lu, A. Rajeswaran, K. Lee, A. Grover, M. Laskin, P. Abbeel, A. Srinivas, I. Mordatch. *Decision Transformer: Reinforcement Learning via Sequence Modeling.* NeurIPS 2021, arXiv:2106.01345.

Summary. Casts offline RL as conditional sequence modelling. A GPT-style transformer is trained to predict next actions given a window of (R, s, a) tokens. At test time the user supplies a target return; the model autoregresses. Matches or beats CQL on D4RL with no Bellman backup.

Why it matters. Removed the assumption that "RL needs TD". Cleared the path for transformer-scale offline policies (Q-Transformer, RT-2, Gato).

> "Despite its simplicity, Decision Transformer achieves state-of-the-art performance on Atari, OpenAI Gym, and Key-to-Door tasks."

Commentary. The simplicity is the point. There is no value bootstrap, no out-of-support sampling, no policy KL constraint — just supervised learning with a clever conditioning signal.

2 Janner et al. (2021) — Trajectory Transformer

M. Janner, Q. Li, S. Levine. *Offline RL as One Big Sequence Modeling Problem.* NeurIPS 2021, arXiv:2106.02039.

Summary. Discretises state, action, reward to a single token vocabulary and applies beam search over multi-step continuations. Plans by generation: the model proposes many trajectories, returns are accumulated, the best continuation is selected.

Why it matters. Bridge between DT (no planning) and MuZero (search-over-learned-dynamics). TT is in practice slower than DT but more robust on long-horizon tasks.

> "We find that beam search over the full trajectory distribution yields qualitatively different behaviour from token-by-token argmax decoding."

Commentary. A pre-figuring of the now-standard "inference-time compute" lesson: planning at inference can substitute for value learning.

3 Janner et al. (2022) — Diffuser

M. Janner, Y. Du, J. Tenenbaum, S. Levine. *Planning with Diffusion for Flexible Behavior Synthesis.* ICML 2022, arXiv:2205.09991.

Summary. Train a diffusion model over (s, a) trajectories with classifier-free guidance. At test time, samples are guided by a value function (or any differentiable cost). Generates entire plans in one denoising pass; supports flexible conditioning.
Why it matters. First paper to show diffusion can be a competitive planning method. Pre-cursor to DIAMOND world models (week 16).

> "Diffusion provides a natural mechanism for sample-then-rerank planning: we can sample many trajectories cheaply and re-rank by any downstream objective."

4 Chi et al. (2023) — Diffusion Policy

C. Chi, S. Feng, Y. Du, Z. Xu, E. Cousineau, B. Burchfiel, S. Song. *Diffusion Policy: Visuomotor Policy Learning via Action Diffusion.* RSS 2023, arXiv:2303.04137.
Summary. A conditional denoising diffusion model over short action chunks, conditioned on a window of image observations. Decisively outperforms BC-RNN and BC-Transformer on contact-rich manipulation tasks. Effectively the new standard for visuomotor BC.
Why it matters. Demonstrates that for *multi-modal* action distributions (which appear in nearly all real-world demonstration datasets), Gaussian or categorical heads fail and diffusion succeeds.

> "Diffusion Policy gracefully handles multi-modal action distributions and is robust to temporal aliasing."

Commentary. The multi-modality observation is the deep one: human demonstrations contain different conventions for the same task, and a unimodal head averages them into a non-functional policy.

5 Hansen-Estruch et al. (2023) — IDQL

P. Hansen-Estruch, I. Kostrikov, M. Janner, J. Grudzien, S. Levine. *IDQL: Implicit Q-Learning as an Actor-Critic Method with Diffusion Policies.* arXiv:2304.10573.
Summary. Train an IQL value model and use it to weight or re-rank samples from a diffusion behaviour clone. Achieves SOTA on D4RL with a small training-time cost.
Why it matters. Cleanly separates "what is plausible under the data" (diffusion BC) from "what is good" (IQL value). The same factorisation now underlies most production offline pipelines.

Sourced from `research/offline_modelbased_sota.md` (sections A.8–A.13).